

# Moist Carrot Cake

Macerated + pressed carrot. Toasted-walnut butter distributed through the crumb. Mascarpone-tempered cream cheese frosting that holds at galley temperature.

Yield: **12 portions** (9×13" sheet)    Active: **25 min**    Total: **1 hr 40 min**    Oven: **175°C** (350°F)    Done: **95°C** internal

|                                                                                                                                                                                                                           |                                                                                                                                                                              |                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WHERE IT WAS BORN</b><br>Medieval Europe, where sweet roots stood in for costly sugar — an English “pudding in a Carret root” appears by 1591; revived in WWII Britain under the Ministry of Food’s <i>Dr Carrot</i> . | <b>EATEN, AUTHENTICALLY</b><br>The ration-table cake: naturally sweetened with carrot when sugar ran short, eaten plain or sugar-dusted as a teacake — no frosting in sight. | <b>HOW IT’S USED NOW</b><br>An American classic since the 1960s health-food revival; the cream-cheese frosting (Philadelphia, 1965) is its modern signature. |
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## INGREDIENTS

METRIC · US VOLUME IN RIGHT COLUMN

### MACERATION

|                                             |                  |
|---------------------------------------------|------------------|
| Carrots, peeled, grated                     | 14.1 oz · ~5 med |
| Light brown sugar ( <i>for macerating</i> ) | 1.1 oz · 2 tbsp  |
| Fine sea salt ( <i>draw moisture</i> )      | 0.11 oz · ½ tsp  |

### DRY

|                      |                  |
|----------------------|------------------|
| All-purpose flour    | 9.2 oz · 2¼ cups |
| Baking soda          | 0.21 oz · 1 tsp  |
| Baking powder        | 0.18 oz · 1¼ tsp |
| Cinnamon, ground     | 0.18 oz · 2 tsp  |
| Ginger, ground       | 0.11 oz · 1 tsp  |
| Nutmeg, fresh-grated | 0.04 oz · ⅓ tsp  |
| Diamond kosher salt  | 0.18 oz · 1 tsp  |

### WET

|                                              |                      |
|----------------------------------------------|----------------------|
| Light brown sugar                            | 6.7 oz · scant 1 cup |
| Granulated sugar                             | 3.5 oz · ½ cup       |
| Neutral oil ( <i>grapeseed/sunflower</i> )   | 7.1 oz · scant 1 cup |
| Eggs, large ( <i>RT</i> )                    | 5.3 oz · ×3          |
| Pure vanilla extract                         | 0.28 oz · 1 tsp      |
| Toasted-walnut butter ( <i>sub-recipe</i> ↓) | 2.1 oz · 3 tbsp      |
| Orange zest, microplaned                     | 0.14 oz · 1 orange   |

### FROSTING

|                                        |                  |
|----------------------------------------|------------------|
| Cream cheese, full-fat ( <i>cold</i> ) | 12 oz            |
| Mascarpone ( <i>cold</i> )             | 4.1 oz · ½ cup   |
| Unsalted butter, soft ( <i>20°C</i> )  | 3 oz · 6 tbsp    |
| Icing sugar, sifted                    | 9.9 oz · 2¼ cups |
| Pure vanilla extract                   | 0.18 oz · 1 tsp  |

Kit: 23×33 cm (9×13") pan · food processor · colander + towel · scale · thermometer · stand mixer · offset spatula · parchment

## METHOD

- Macerate carrot.** Toss 400g grated carrot with 30g brown sugar + 3g salt. Colander over bowl. Stand **20 min**.

*Osmosis draws ~30% free water out of carrot cells → concentrated flavour, less soggy crumb.* — McGee 2004
- Press dry.** Squeeze in clean towel. Target ≤280g pressed carrot. Reserve 40g liquid (optional glaze).

**SUB-RECIPE — TOASTED WALNUT BUTTER**  
**Raw walnut halves, 160g** · pinch fine salt (optional) · *yields ~70g butter + 80g chopped garnish*  
**2a.** Toast all 160g at **165°C (325°F)**, 10–12 min. Cool 5 min.  
**3a.** Split: **80g → food processor** 3–4 min → smooth butter (~70g). Use 60g in batter; 10g extra keeps 2 wk fridge.  
**4c.** **Remaining 80g:** rough chop for garnish.  
**5d.** *Maillard furaneol + pyrazines in the fat phase → roasted note in every bite.* — López-Alt / Modernist
- Preheat 175°C** (350°F). Butter + parchment 9×13" pan. Whisk all dry together.
- Whisk wet.** 190g brown sugar, 100g granulated, 200g oil, 150g eggs, vanilla, 60g walnut butter, orange zest. 1 min smooth.
- Fold carrot into wet.** 4 folds.
- Add dry in 2 additions.** Half → 6 strokes → second half → fold until just incorporated. Flour streaks fine. Tap pan twice. Bake **28–34 min**.

*Soda reacts with brown sugar acidity immediately; powder activates on heat. Sequential lift prevents collapse before crumb sets.* — McGee 2004

**Done:** Internal **95°C** (203°F). Top springs back. Edges pull from pan. Toothpick: moist crumbs only.  
**Cool:** 15 min in pan → wire rack. Frost only at ≤22°C (at least 60 min).

**Frosting** — mix while cake cools

- 6a.** Paddle **cold cream cheese** 30 sec only. Add cold mascarpone, beat 20 sec.
- 7a.** Add softened butter, beat 60 sec. Sift icing sugar in 2 additions, mix to incorporate only.
- 8c.** Vanilla + pinch salt. Taste. If slack: refrigerate 15 min. *Mascarpone (60–75% fat) buffers frosting stability at 28°C.* — ATK 2021
- 9d.** **Frost:** Offset spatula, 1 cm layer. Scatter toasted walnuts + orange zest. Chill 30 min before slicing.

### ONE BATTER, TWO DISHES — SPICED CARROT TEA-LOAF

The frosting-free original. Pour the same batter into **two 900 g (2 lb) loaf tins**, two-thirds full; bake **175°C, 45–55 min** to 95°C internal (foil at 35 min if browning). No frosting — glaze with the reserved carrot liquid + sugar, or leave plain. *Keeps 2–3 days at room temp, no cold chain — the smart passage bake and the crew tier.*

## ELEVATION

### TIER 1 — NO EXTRA TIME

- Cardamom + white pepper** — 1g + 0.5g in dry
- Miso in frosting** — 10g white miso with butter
- Dark brown sugar swap** — more molasses depth

### TIER 2 — WORTH IT (+15 MIN)

- Carrot-liquid glaze** — reduce 40g liquid + 20g sugar, brush warm cake
- Candied ginger** — 50g diced, fold in at end
- Part brown butter** — replace 100g oil with 85g browned

### TIER 3 — RESTAURANT

- Dehydrated carrot chips** — mandoline + 65°C / 2 hr, press into frosting
- Pineapple soak** — 40g cold juice, brush warm cake
- Toasted coconut** — 40g golden, press into fresh frosting

## TROUBLESHOOT

|                           |                                                                           |
|---------------------------|---------------------------------------------------------------------------|
| <b>Dense, wet crumb</b>   | Carrot not pressed. Target ≤280g. Squeeze in towel.                       |
| <b>Frosting weeps</b>     | Cream cheese too warm or over-beaten. Start cold. 30 sec max on CC alone. |
| <b>Sunk centre</b>        | Underbaked. Hit 95°C internal. Lower oven to 165°C if recurring.          |
| <b>Rubbery crumb</b>      | Overmixed. Count 12 strokes max after dry goes in.                        |
| <b>Frosting too sweet</b> | Add 10g cold cream cheese, mix 10 sec. Or pinch more salt.                |
| <b>Muddy spice</b>        | Stale spices. Replace if >6 months old. Freshly grate nutmeg always.      |

## CHARTER PREP & STORAGE

|                         |                                                                     |
|-------------------------|---------------------------------------------------------------------|
| <b>Freeze unfrosted</b> | <b>Always. Cream cheese frosting breaks on thaw.</b>                |
| <b>Naked cake</b>       | Cling + foil. 2 mo freezer. Thaw overnight in fridge.               |
| <b>Frosted cake</b>     | 3 d fridge. Cover loosely — no cling on frosting.                   |
| <b>Walnut butter</b>    | 2 wk fridge, jar. Stir if oil separates.                            |
| <b>Pressed carrot</b>   | 3 d fridge, airtight. Drain before using.                           |
| <b>Scaling</b>          | ×2 = two pans, same time. 8" rounds: 30–35 min. Muffins: 20–24 min. |